Study of a 16-QAM Conversion Receiver operating at 4. 28 GHz

RF SYSTEMS – 2026/27

General Specifications

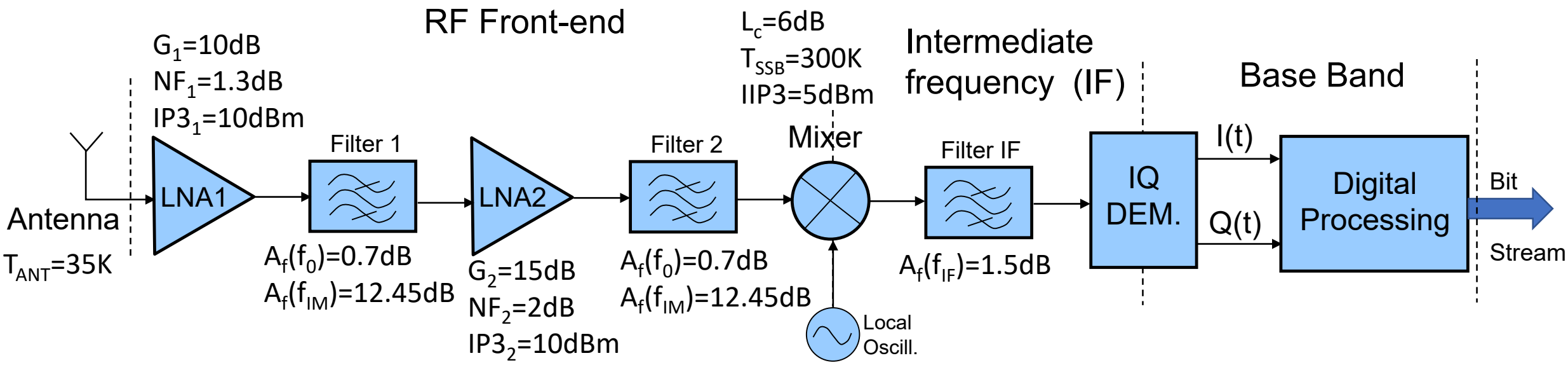
- RF frequency: 4280 MHz;
- IF Frequency: 140 MHz (L.O. 4140 MHz)
- IM frequency: 4000 MHz
- Modulated Signal: 16-QAM
- Symbol Rate: S=30 Msymbol/sec.
- alfa=0.35

Signal bandwidth: B=30*(1+alfa)=40.5 MHz

Data Rate: $R=S*\log_2(16)=120$ Mbit/sec

Goal: BER=1e-3 with minimum Input Power

Receiver configuration

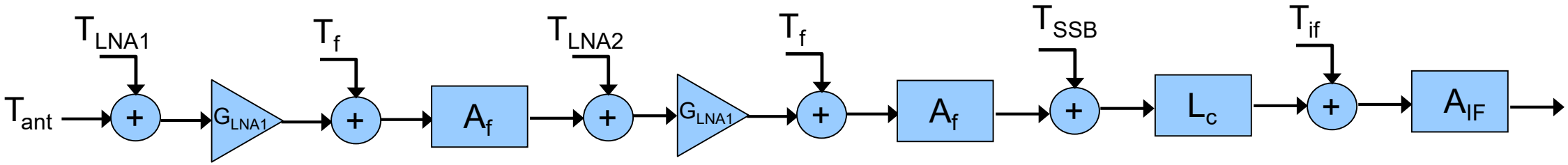


Note: VSS requires to specify the mixer noise by
NFSSB=10*log10(TSSB/T0+2)=4.82 dB (T0=290 K)

IQ Dem. + BB process.= Ideal (no noise contribution)

Evaluation of T_{eq}

Discarding contribution from IM frequency (filter attenuation at IM = infinity):



$$T_{eq} = T_{ant} + T_{LNA1} + \frac{T_f}{G_{LNA1_lin}} + \frac{T_{LNA2} \cdot A_{f_lin}}{G_{LNA1_lin}} + \frac{T_f \cdot A_{f_lin}}{G_{LNA1_lin} G_{LNA2_lin}} + \frac{T_{SSB} \cdot A_{f_lin}^2}{G_{LNA1_lin} G_{LNA2_lin}} + \frac{T_{if} \cdot L_c \cdot A_{f_lin}^2}{G_{LNA1_lin} G_{LNA2_lin}}$$

$$= 164.3K$$

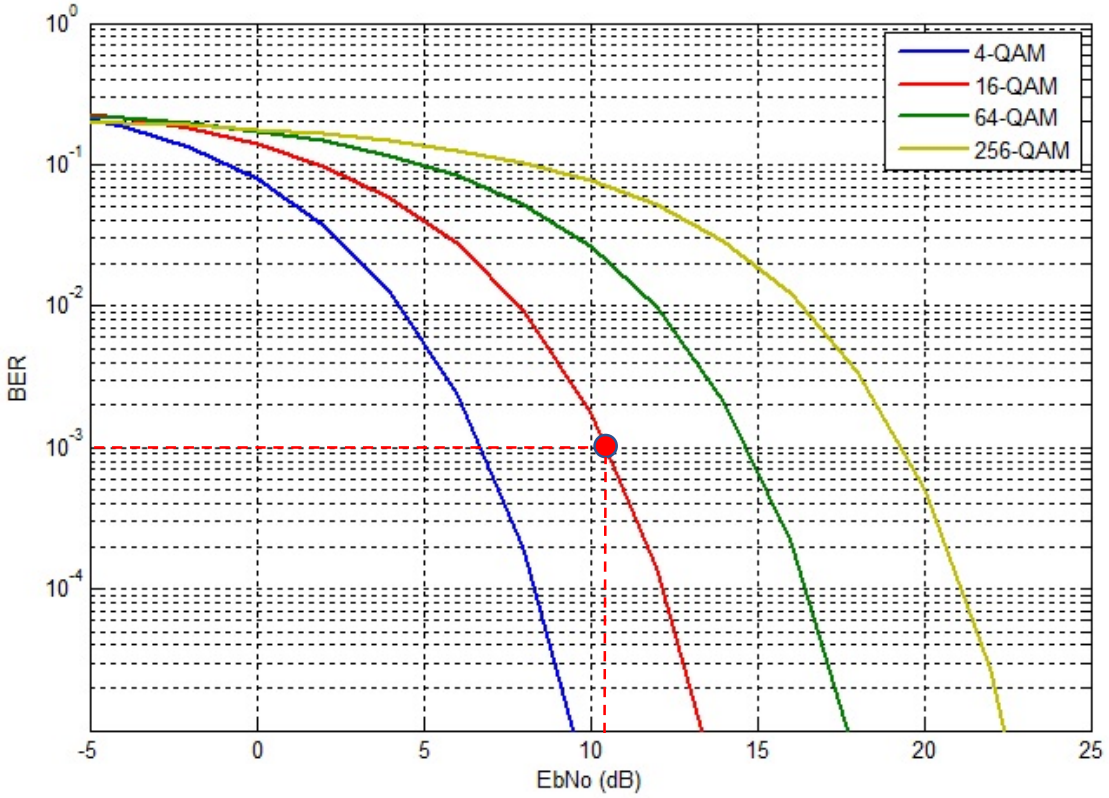
The contributions from IM is equal to 1.9K

Equivalent noise power density at input: $N_{eq}=K \cdot T_{eq}$=-176.4 dBm/Hz

Noise power density at IF end: $N_{IF}=N_{eq}+G_{LNA1}+G_{LNA2}-2 \cdot A_f-L_c-A_{IF}$=-160.3 dBm/Hz

Required SNR and P_{r_min}

M-QAM modulation imposes the following constraint between BER and Eb/N0:



For M=16 and BER=10^{-3} it has:

Eb/N0≈11 dB

Then:

SNR=(D/B)(Eb/N0)=4.71+11=15.7 dB

Being SNR=$P_r/(K \cdot T_{eq} \cdot B)$ we get:

P_{r_min}=-84.6 dBm

Simulation of the receiver with VSS: sampling frequency

- Simulation are carried out on the envelope signal
- To select the sampling frequency (f_s), it must be considered the following constraints (f_c carrier frequency):

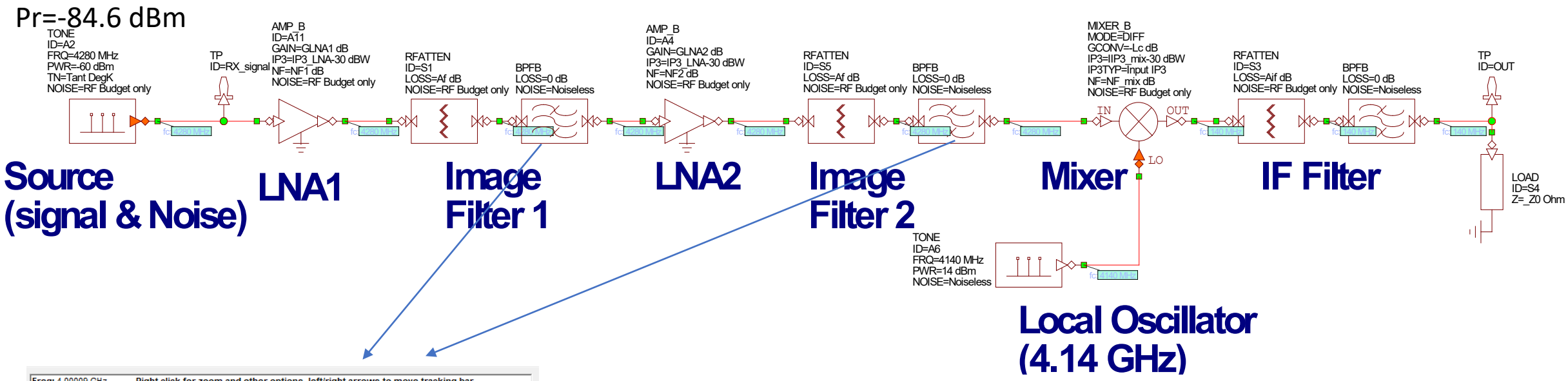
$$2f_c > f_s > 2\left(f_{\max} - f_c\right)$$

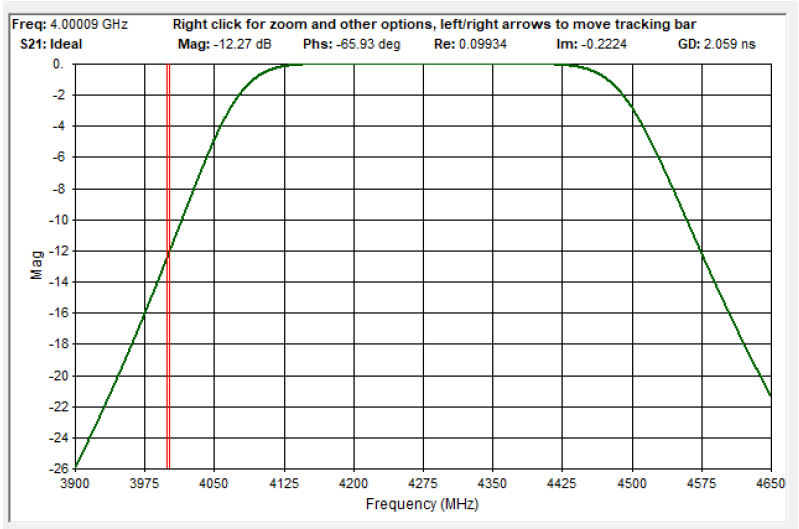
- with f_{max}-f_c=$B/2$.
- For the RF signals we have: f_c=4280 MHz and B=40.5 MHz, so: 8560>f_s>81
- However, at IF it has f_c =f_{IF}=140 MHz. Then: 280>f_s>81
- We then assign: ***f_s*=270 MHz**
- The oversampling factor (OV) is determined by the symbol rate S= f_s /OV=30 Ms/sec → **OV=9**
- The simulation frequency range is (f_c-f_s/2)<f<(f_c+f_s/2) → **4145 < *f* < 4410**.
  It can be observed that the image frequency (4000 MHz) is not included. We need to increase f_s (it will be discussed in the following)

Budget Simulation

- This type of simulation allows to study how the parameters assigned to each block affect the various figure of merits of the receiver (T_{eq}, *IIP3, SNR*, etc.)
- No time domain computations are carried out (parameters of the blocks are propagated along the block diagram representing the receiver)
- It is possible to observe the value of each figure of merit inside the blocks in cascade (at input or output of the blocks)
- This analysis requires a signal source which includes noise
- The simulations is carried out up to the end of IF (the IQ demodulator and BB processing are not included)

Schematic for the Budget simulation





Filter characteristic (selected to get 12.5 dB of rejection at IM)

Note that the filter block is assumed lossless, and an attenuator is added in front of it to add the passband losses (and the noise contribution)

Evaluation of Teq:

Budget_System_Temperature

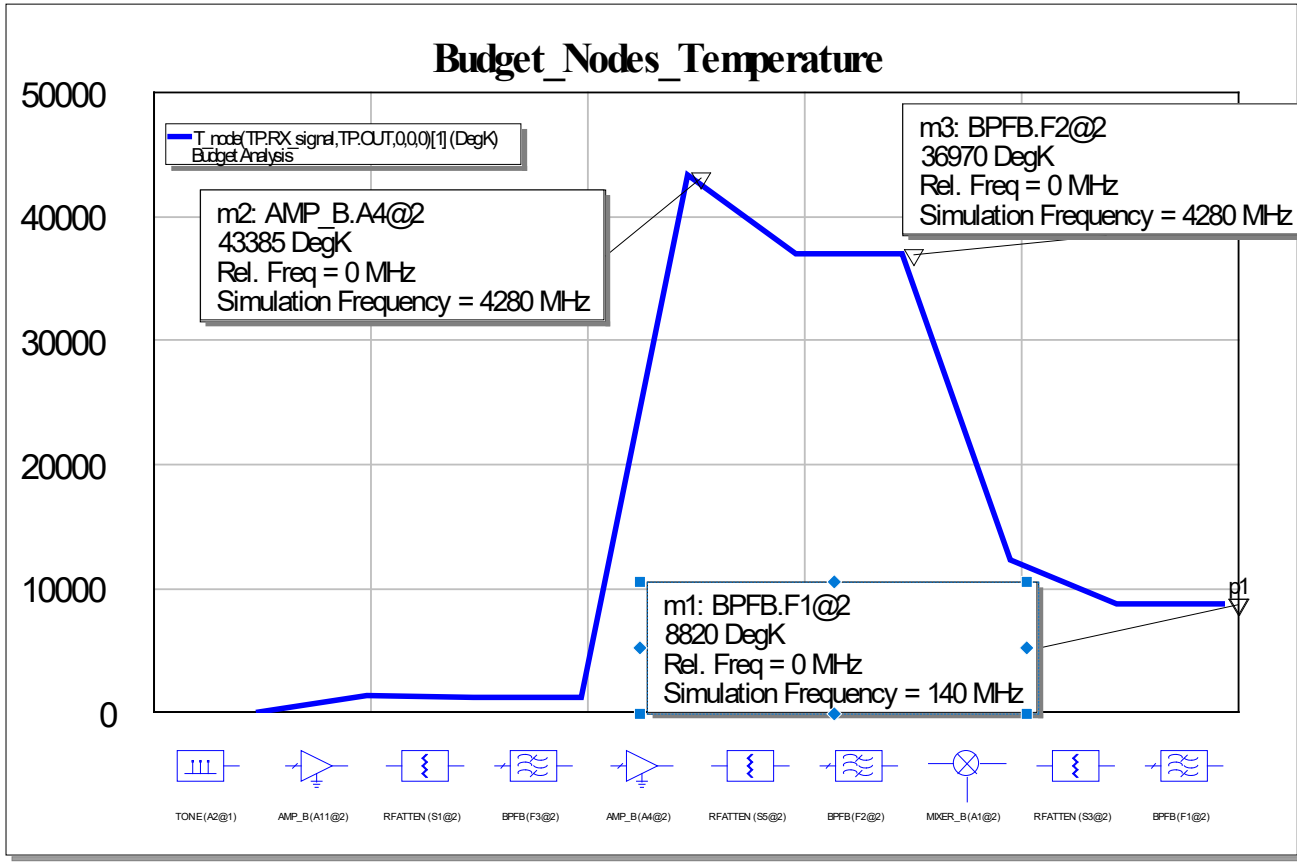
Tsys(TP.RX_signa... Budget Analysis Frequency	Tsys(TP.RX_signa... Budget Analysis System Equivalen... Rel. Freq = 0 MHz
0	164.27

Computed with the formula: 164.3

Cascade Nodes Temperature

At IF end the analytic computation gives:

$T_{IF}=T_{eq}*(G_{LNA1}*G_{LNA2}/(Af^2*L_c*A_{if})=$
$=8822$ K



Cascade Noise Spectral Density at the nodes

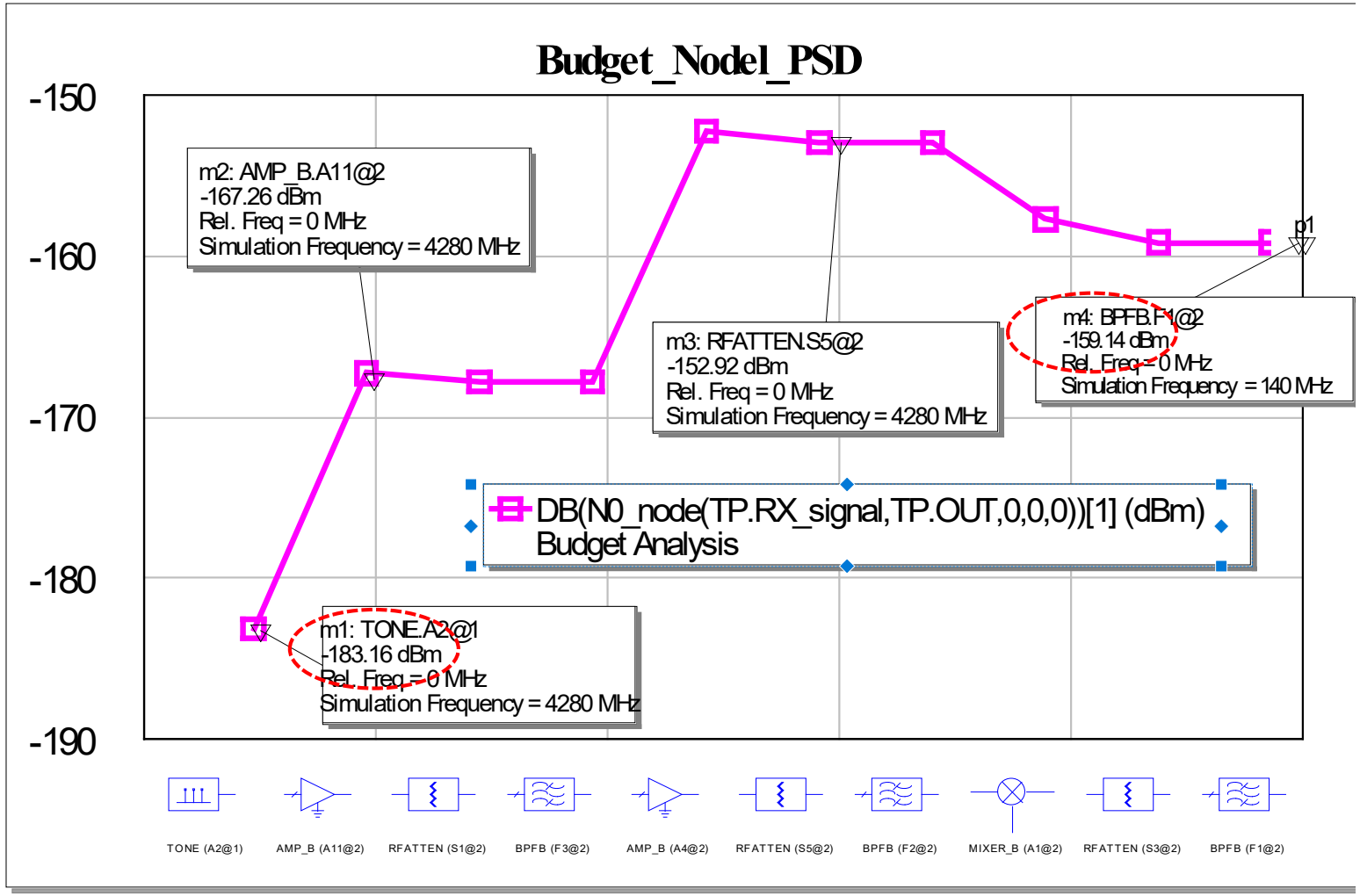
$N=10*\log10(K\cdot T)+30$ dBm/Hz

Input (T_{ant}=35 K):

$N=N_{ant}=K*T_{ant}$=-183.2 dBm/Hz

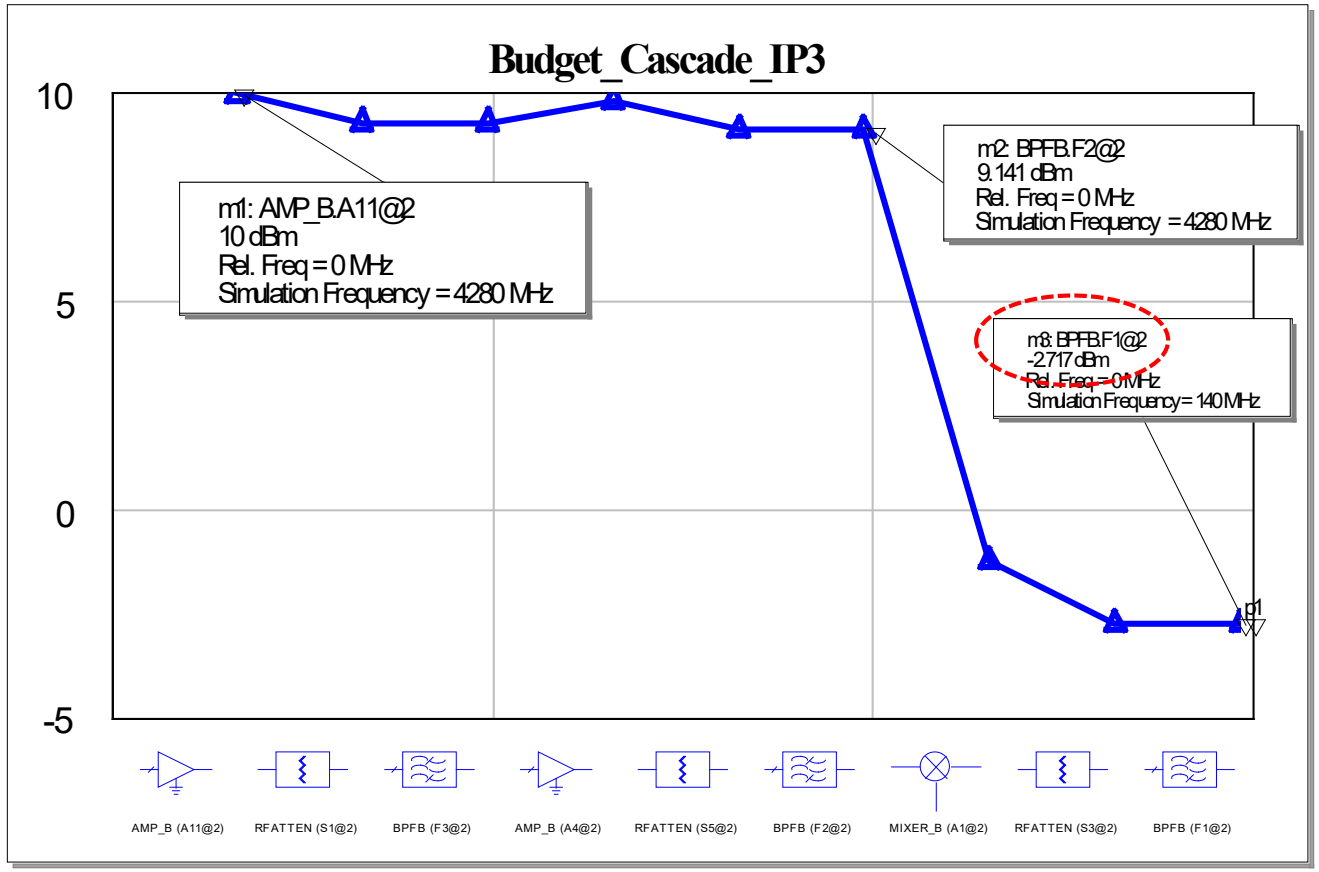
Output (T_{out}=8820 K):

$N=N_{out}=K*T_{out}$=-159.1 dBm/Hz



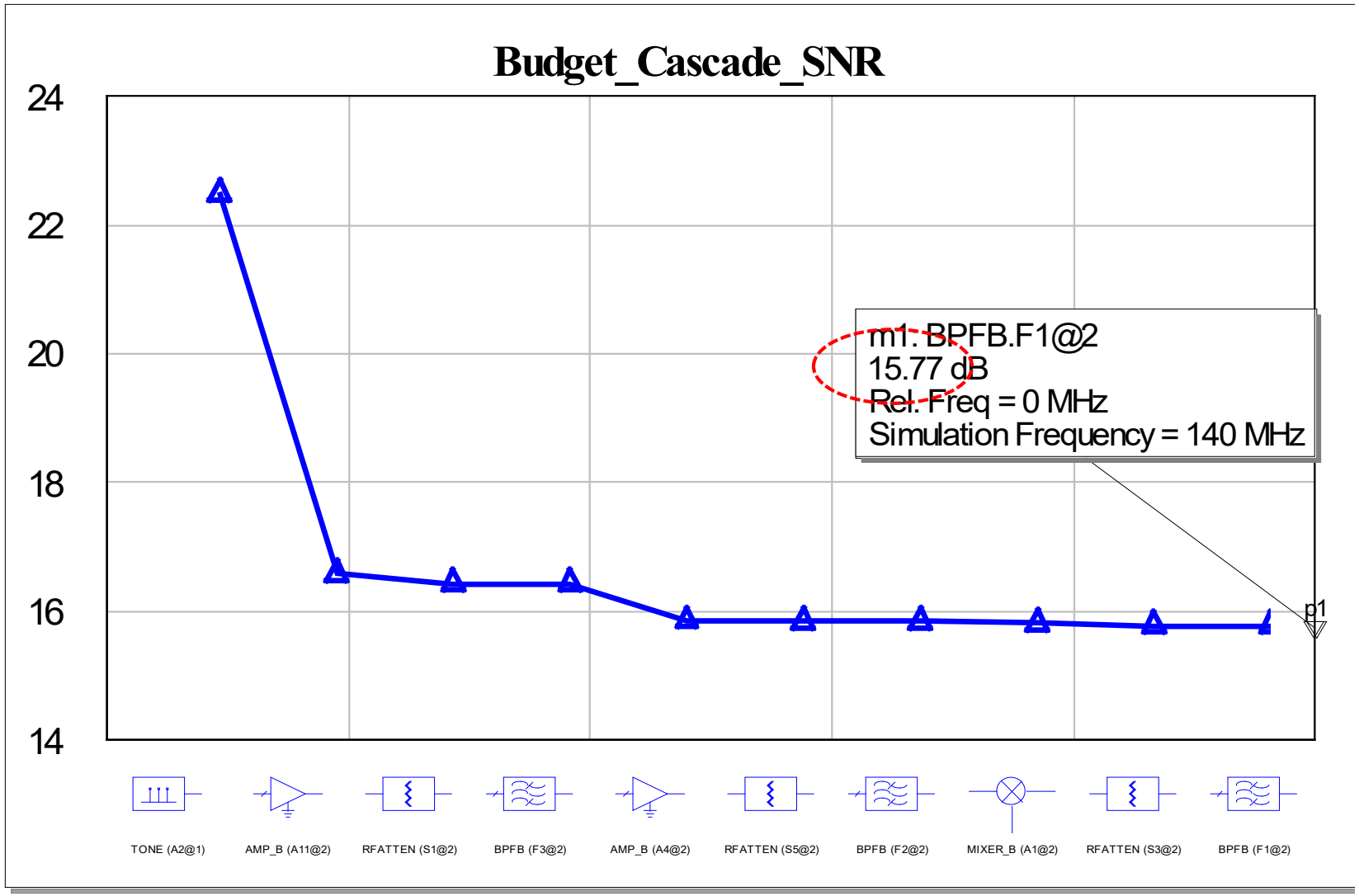
Cascade Intercept Point at the nodes (block output)

$$IIP_{3,eq} = \left(G_{LNA1} \left(\frac{1}{IP3_{LNA}} + G_{LNA2} \left(\frac{1}{A_f \cdot IP3_{LNA}} + \frac{1}{A_f^2 \cdot IIP3_{MIX}} \right) \right) \right)^{-1} = -20.02 \text{ dBm}$$



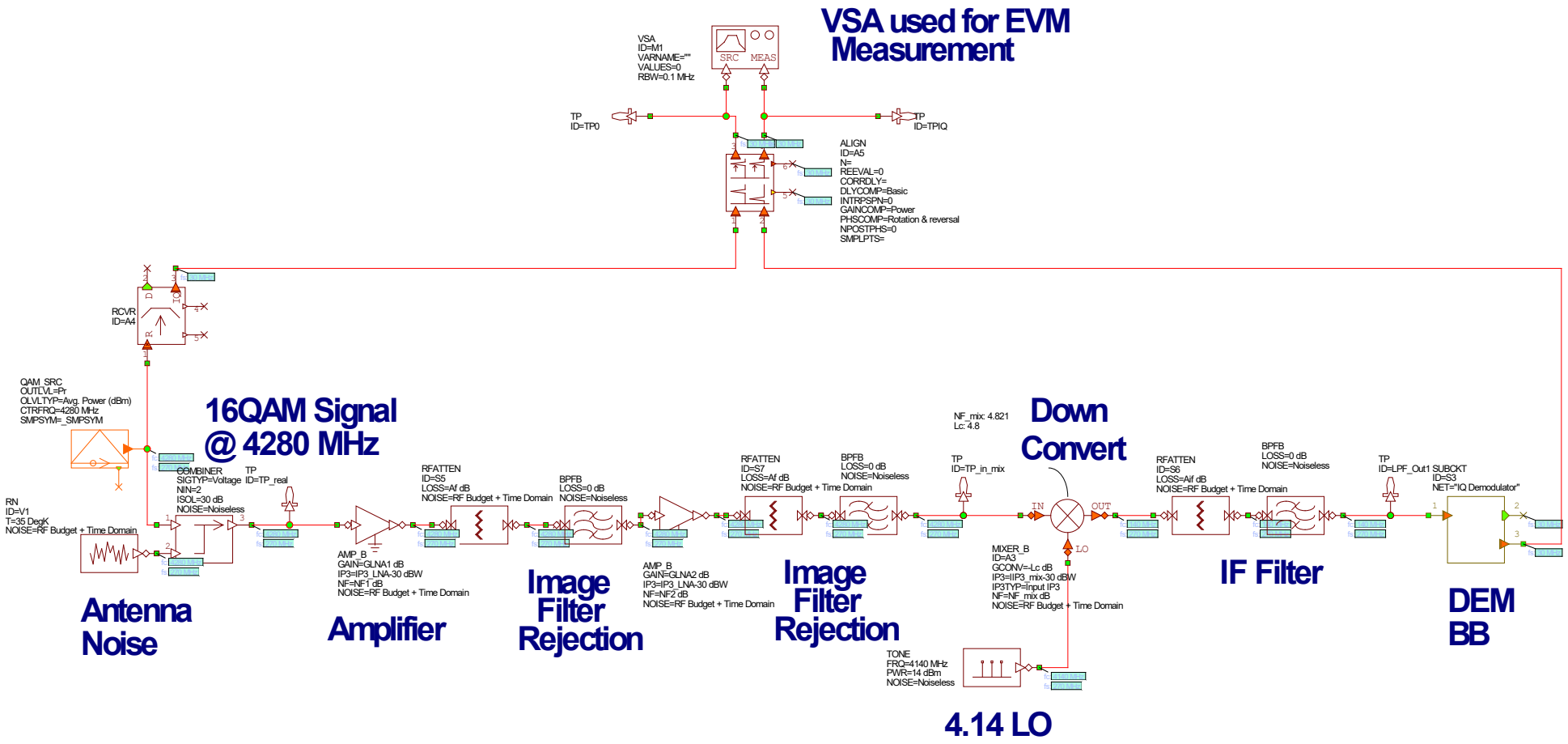
$IP_{3,OUT} = IIP_{3,eq} * G_{LNA1} * G_{LNA2} / (Af^2 * L_c * A_{if}) =$
$= -2.76$ dBm

Cascade SNR (Pr=-84.6 dBm)

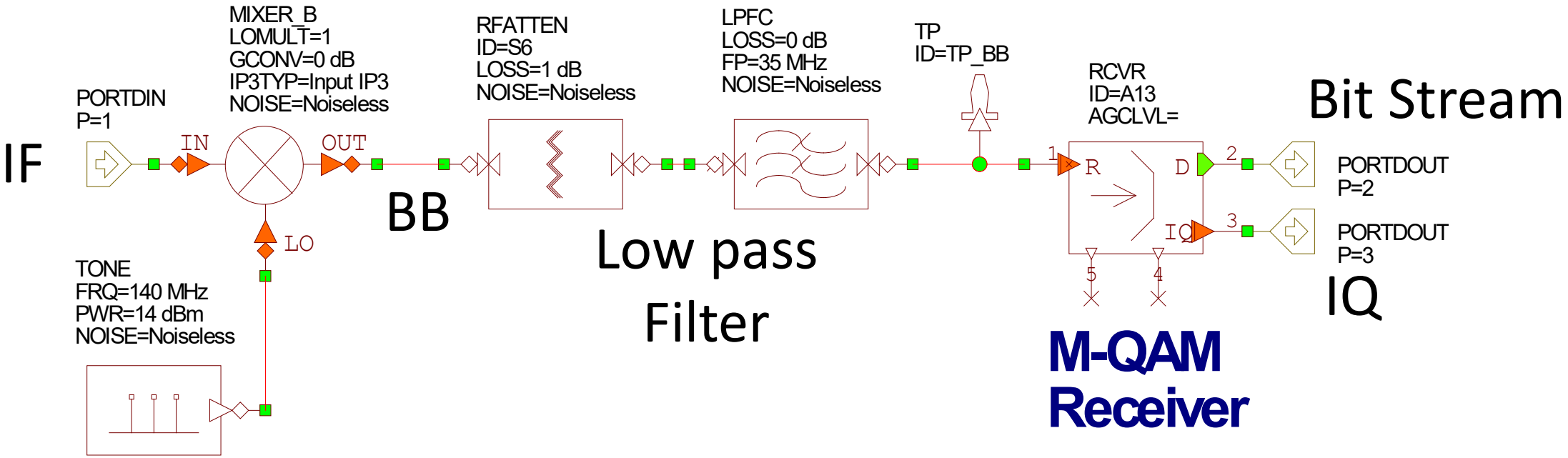


SNR=(D/B)(Eb/N0)=15.7 dB

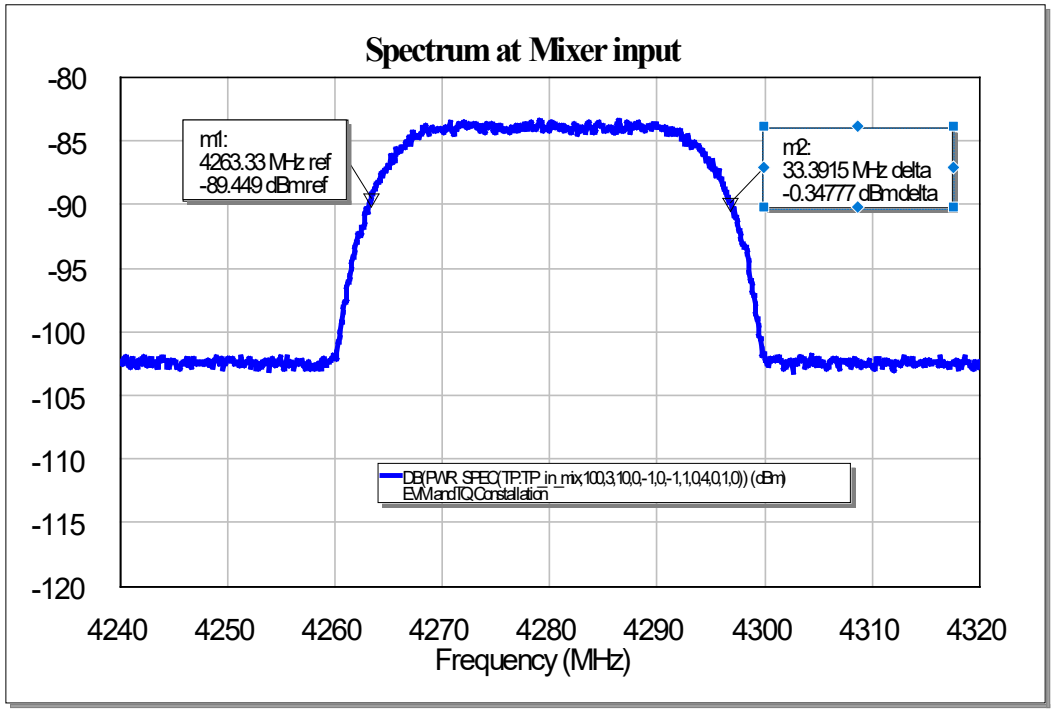
Evaluation of Constellation Diagram and EVM

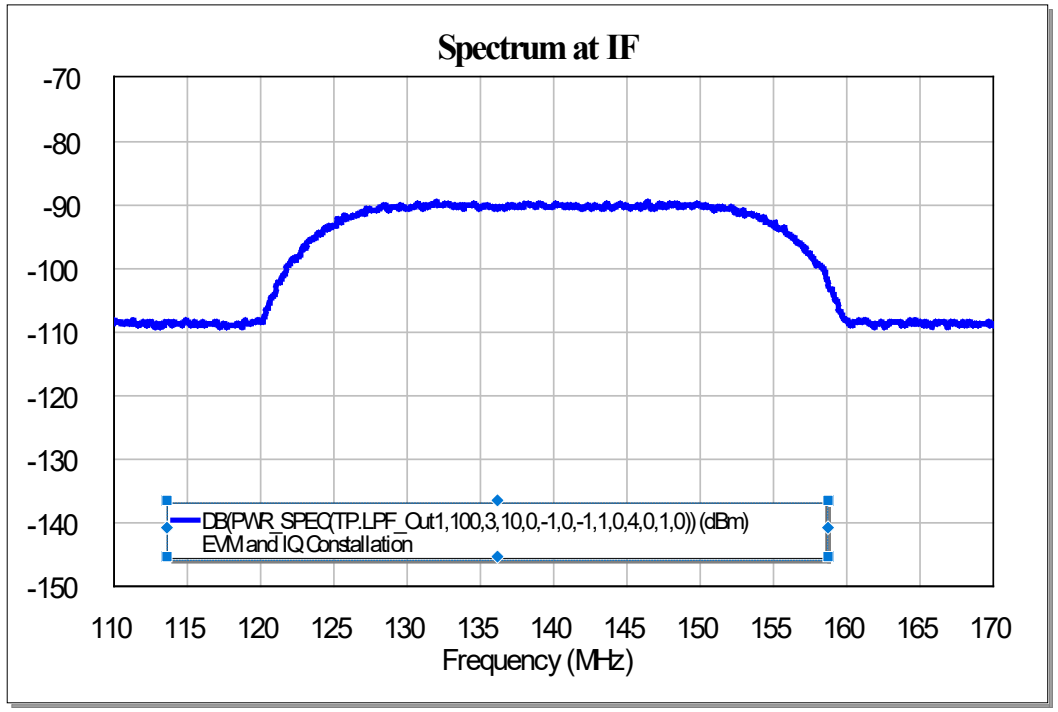


Demodulator and bit stream extraction (lossless)

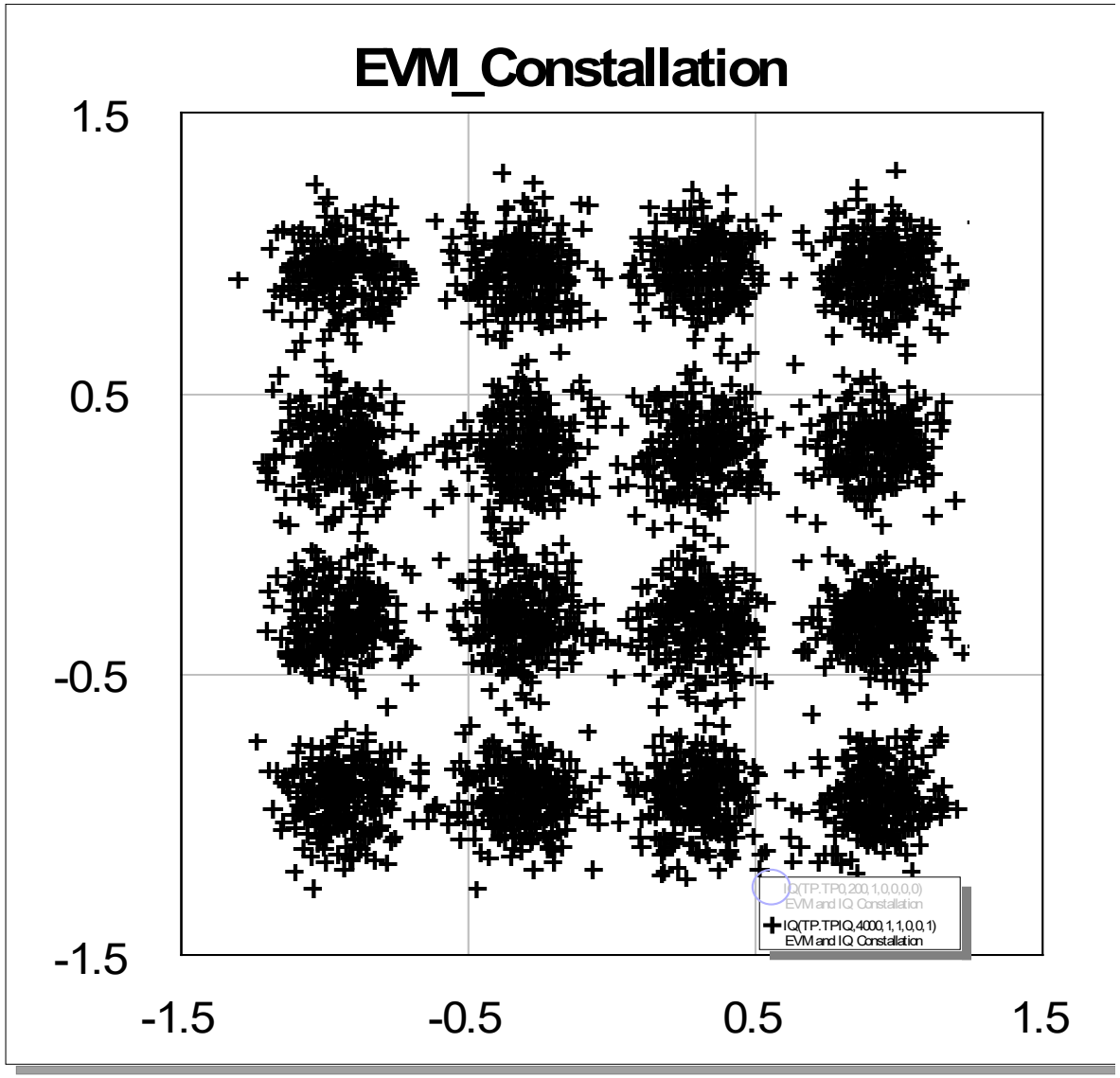


Spectra


Spectrum at Mixer input
-80
-85
-90
-95
-100
-105
-110
-115
-120
4240
4250
4260
4270
4280
4290
4300
4310
4320
Frequency (MHz)
m1:
4263.33 MHz ref
-89.449 dBm ref
m2:
33.3915 MHz delta
-0.34777 dBm delta
DB(PWR_SPEC(TP.TP_in_mix,100,3,10,0,-1,0,-1,1,0,4,0,1,0)) (dBm)
EVM and IQ Constallation

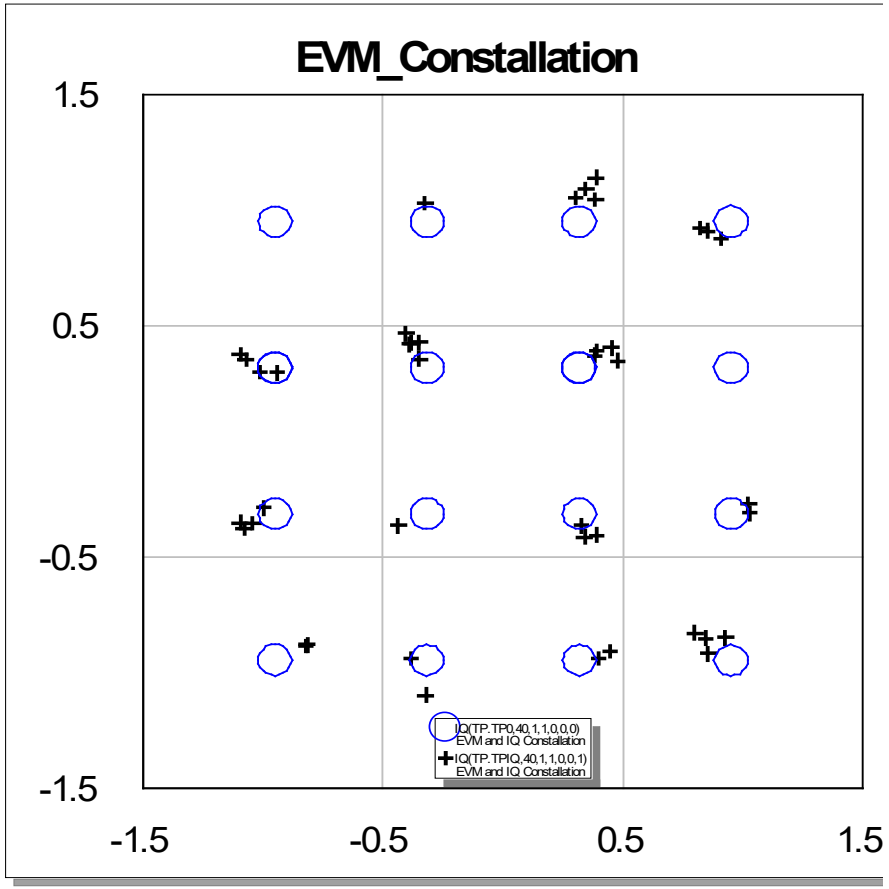

Spectrum at IF
-70
-80
-90
-100
-110
-120
-130
-140
-150
110
115
120
125
130
135
140
145
150
155
160
165
170
Frequency (MHz)
DB(PWR_SPEC(TP.LPF_Out1,100,3,10,0,-1,0,-1,1,0,4,0,1,0)) (dBm)
EVM and IQ Constallation

EVM & Constellation Diagram



Power(dBm)	EVM (%)
-85.2	15

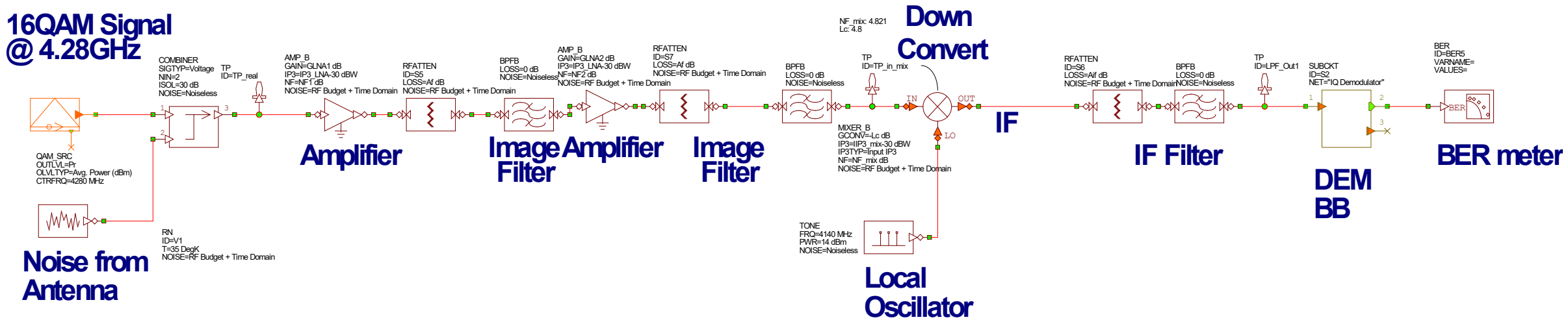
EVM & Constellation Diagram (Pr=-24.2)



Power(dBm)	EVM (%)
-24.2	15

Dynamic Range=-24.2+85.2=61 dB

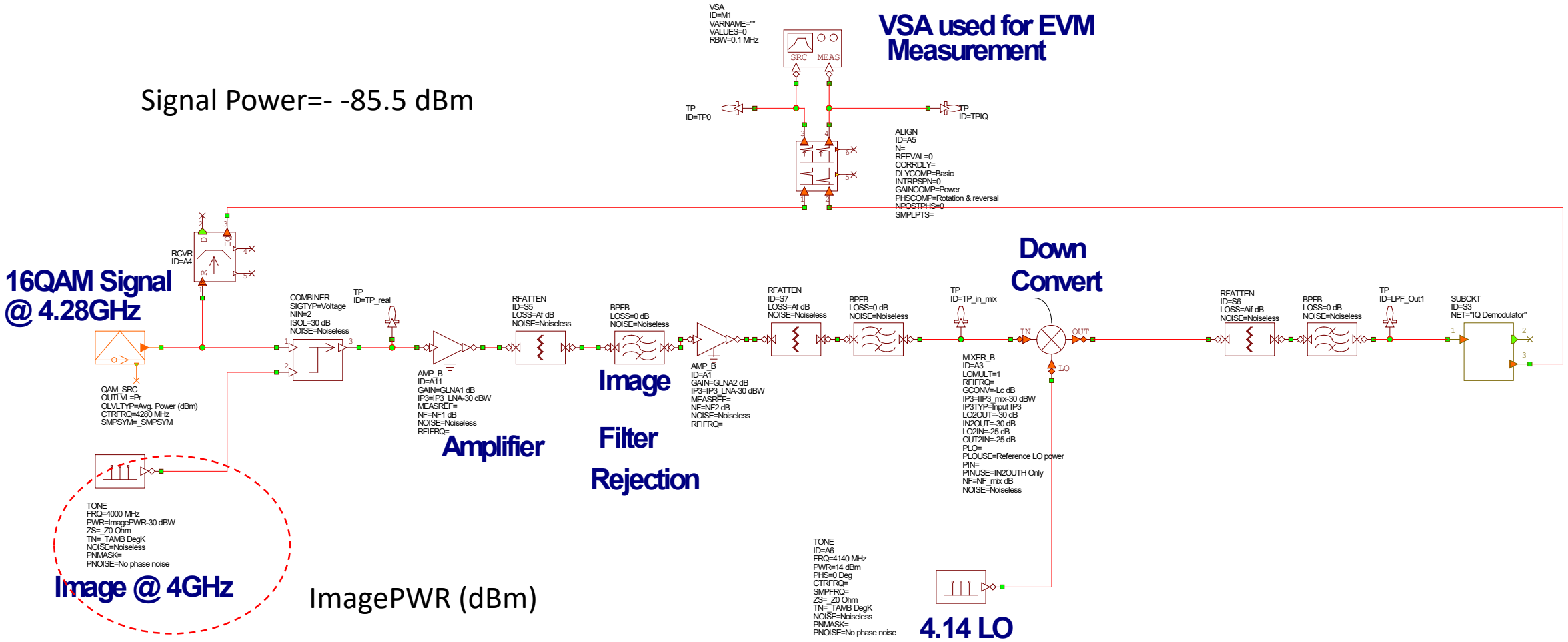
BER Measurement



Power in	BER	DB(EsN0)
-84.2	0.001	10.448
-24.2	0.0014	
-25.4	0.001	

To measure BER <1e-4 would require a very long time!

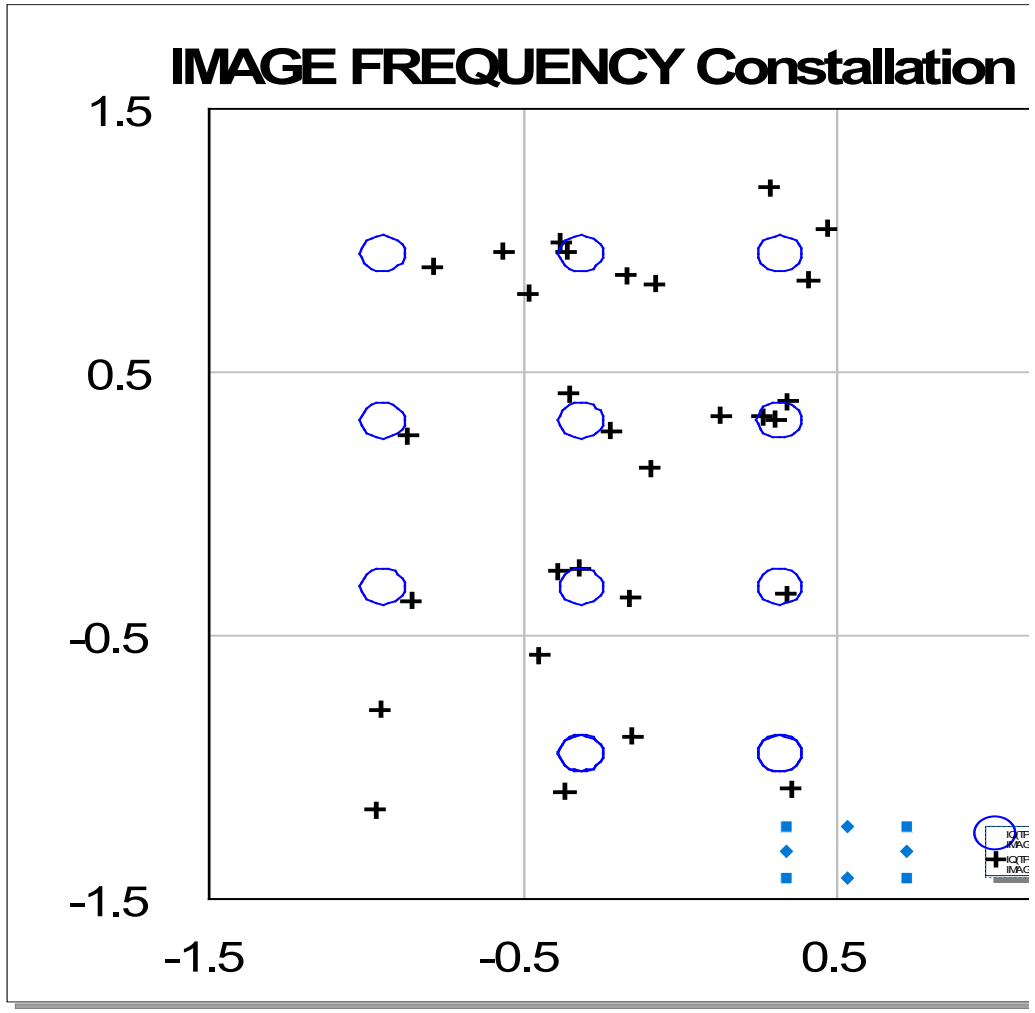
Effect of an interferer at the image frequency



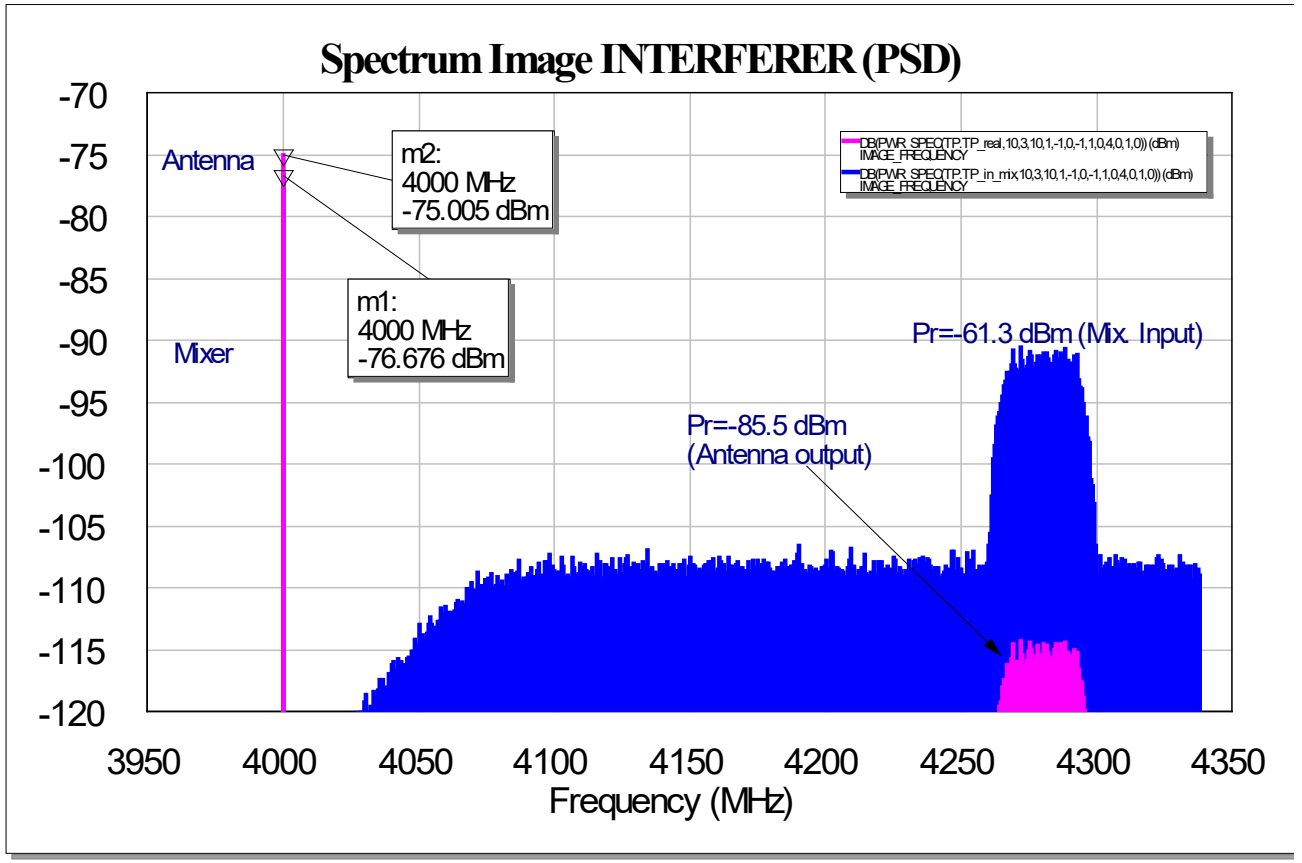
- The sampling frequency must be increased to 900 MHz so that f_{IM} is inside the range of admissible frequencies (fc-fs/2<f<fc+fs/2 → 3830<f<4730).
- At intermediate frequency (140 MHz) it has instead -310<*f*<590. Considering the signal band (40.5 MHz), there are no negative frequencies to analyze at the intermediate frequency (119.75<*f*<160.25).
- However, noise could extend beyond this band but the filter before the mixer prevents this.

EVM & Constellation Diagram: Pr=-84.5 dBm, Image=-75 dBm

EVM=22.9%



RF signal at mix input: -60.9 dBm
Image at mix input: -73.6
SNR=-60.9+73.6=12.7 dB (thermal noise is negligible)



Spectrum at IF with the interferer at f_{IM}

